

From Static to Dynamic: Enhancing AAC with Generative Imagery and Zero-Shot TTS

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Abstract

This paper presents an Augmentative and Alternative Communication (AAC) approach for minimally verbal children with Autism Spectrum Disorder. Traditional AAC systems use fixed symbol sets and pre-defined Text-to-Speech (TTS) voices, this proposed method leverages text-to-image generation and zero-shot TTS to expand expressive capabilities. Users can create visual symbols for concepts and interests, enabling richer communication.

Further, zero-shot TTS allows users to upload or record personalized voices, enabling users to have individualized output. By minimizing reliance on static symbols and voices, this approach aims to increase communicative agency, personal relevance, and social validity, areas often neglected in traditional interventions. Future research will explore long-term effects on communicative skills, user satisfaction, social engagement, and adaptability across various cultural and linguistic settings, aiming to develop more dynamic and personalized AAC solutions.

Index Terms: Speech Synthesis, TTS, AAC, Human-Computer Interaction

1. Introduction and Related Work

A considerable proportion of children diagnosed with autism spectrum disorder (ASD), approximately 25% to 30%, experience challenges in functional spoken language development. Many remain minimally verbal, rely on echolalia, or only communicate verbally in limited contexts [1]. For these children, Augmentative and Alternative Communication (AAC) devices, including low-tech systems such as the Picture Exchange Communication System (PECS) and high-tech Speech-Generating Devices (SGDs), often serve as critical supports for communicative development [2].

Research supports the effectiveness of AAC interventions in enhancing communication for children with ASD [3], particularly PECS and SGDs. However, many interventions focus narrowly on requesting rather than broader communicative intents, often represented by available vocabulary on AAC devices [4, 5].

Current AAC systems are often constrained by the limited images and vocabulary within them, which can make it difficult to support the development of richer, more personalized communication. Measures of social validity are frequently limited to surveys of caregivers or educators that assume any improvement in functional communication is inherently socially valid [5]. This reflects a need to prioritize real-world application and user-centered design in AAC research.

Advancements in mobile technology have made robust, affordable AAC devices feasible [6]. However, high costs and necessity for clinician approval for many devices have limited ac-

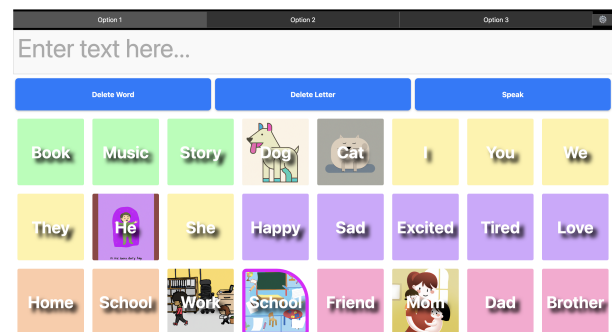


Figure 1: Core interface, showcasing generated images, voice selection, and folders.

cessibility for general users seeking SGDs. Many users perform effectively with SGDs when provided with minimal or no instruction [7], as such low-cost applications could enhance communication by removing barriers. Additionally, text-to-image (T2I) and zero-shot text-to-speech (TTS) technologies enable on-demand creation of visual symbols and voices, providing immediate adaptation to user needs, rather than being restricted via designer-imposed limitations.

2. System Design

2.1. User Interface and Interaction

The system is cross platform and is designed to appear similar to existing picture based AAC systems, to the existing SGD “Proloquo2Go” [8]. This is to ease current users in adapting to the new system, and keeps the system inline with AAC design principles that have previously been effective with actual users [8].

The core section of the display consists of large square buttons that users can press. The buttons feature either an image or a solid color as their background. Each button displays words or phrases. Users can press these to add the displayed text to their next speech output or long-press them to enable options for removal, editing, or relocation of the button to other positions on the board.

For the buttons images, users can either upload local images to the platform or generate images via a T2I system, specifically DALL-E 3 [9]. The buttons ‘word’ (the word added to the output when pressed) is used within a predefined prompt structure for the T2I model. Four image are generated to be selected as the buttons background.

Buttons can function as folders, allowing users to open new

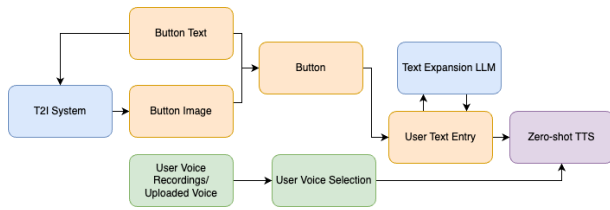


Figure 2: Diagram of the system usage flow.

“pages” that organize words often accessed together. For example, in recreational contexts, words related to specific toys, action phrases like “throw” and “catch” or descriptive terms like “fast” and “bouncy,” can be grouped.

The color of buttons can be customized, with a selection of neutral tones and mellow shades available by default, along with custom color selection also.

To enable communication in a voice ‘fitting’ to them, a zero-shot TTS system is integrated into the platform. Users are able to select the voice they wish to communicate in by selecting options along the top bar of the interface. The voices can be customized through either uploading short audio clips of a voice they wish to use, or by recording their voice. Simple sentences are provided for the user to read out, which are used as input, along with the text to speak to the zero-shot TTS system. XTTS was chosen as the TTS system due to its multilingual capabilities [10], allowing for better accommodation of diverse backgrounds.

2.2. Customization and Accessibility Options

Within the device settings, caregivers or users can tailor the interface to individual needs. They can adjust margins to reduce accidental presses, modify the number of boxes per row, change the long-press duration for entering editing mode, enable or disable text expansion, and lock editing to prevent unintended changes.

2.3. Additional Features

Users or carers can enable ‘text expansion’. This allows the user to create expanded versions of their current input that is potentially more conversationally appropriate. This operates similarly to a previous AAC research prototype [11] and is currently performed using GPT4o, however future work should enhance this with fine-tuned models for each user, fitting to their personal style. Further, prior to this being available outside of an experimental context, local LLM solutions should be utilized to prevent conversational data being shared with third parties. The system is available here: <https://github.com/jandfranc/AIImageAAC>.

2.4. Ethical and Privacy Considerations

The system is designed with modularity to enable adaptation to improving technologies to ensure components like T2I systems, which may be trained on data sourced without permission, can be replaced with alternatives aligning with ethical standards. Similarly, the TTS component is designed to be exchangeable as better methods become available.

To protect AAC device users privacy, personal data is not shared with third-party servers unless absolutely necessary. Local data processing is prioritized to enhance security. Before deployment, extensive testing is essential to ensure safety and

security.

2.5. Future Research

Future research should compare the system’s effectiveness with existing AAC solutions, investigating functional and social-communicative functions. Longitudinal studies should assess whether communication improvements persist and generalize across varied settings. Research should incorporate feedback from AAC users, caregivers, and professionals to meet their needs. Additionally, studies should evaluate the system’s scalability, cross-linguistic applicability, and integration with other assistive technologies to broaden accessibility and usability for diverse populations.

3. References

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